INTERMEDIATE PART-II (12th Class)	2026 (1st-A)	Roll No:
MATHEMATICS PAPER-II GROUP-I		
TIME ALLOWED: 2.30 Hours	SUBJECTIVE	MAXIMUM MARKS: 80

NOTE: Write same question number and its parts number on answer book, as given in the question paper.

SECTION-I

2. Attempt any eight parts. $8 \times 2 = 16$

(i)	Define Identity Function. Give example.		
(ii)	Show that the equations $x = a\sec\theta,\ y = b\tan\theta$ represent the equation of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$		
(iii)	Let $f: R \to R$ be the function, $f(x) = 2x + 1$. Find $f^{-1}(-1)$	(iv)	Evaluate $\lim_{x \to 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$
(v)	Express limit in term of 'e' $\lim_{x \to \infty} \left(1 + \frac{3}{x}\right)^{2x}$	(vi)	Differentiate w.r. to 'x' of $\frac{x^2 + 1}{x^2 - 3}$
(vii)	Find $\frac{dy}{dx}$ if $y(x^2 - 1) = x\sqrt{x^2 + 4}$	(viii)	Differentiate w.r. to variable involved, $x^2 \sec 4x$
(ix)	If $f(x) = \ell n(x + \sqrt{x^2 + 1})$, what is $f'(x)$?	(x)	Calculate y_2, if $x^2 + y^2 + 2gx + 2fy + c = 0$
(xi)	Determine the intervals in which $f(x)$ is increasing or decreasing for the domain mentioned $f(x) = \sin x : x \in (-\pi, \pi)$		
(xii)	The perimeter of a triangle is $20\,cm$. If one side is of length $8\,cm$, what are lengths of the other two sides for maximum area of triangle?		

3. Attempt any eight parts. $8 \times 2 = 16$

(i)	Find dy in $y = \sqrt{x}$ when x changes from 4 to 4.41.	(ii)	Evaluate $\int x(\sqrt{x} + 1)\,dx,\ (x > 0)$
(iii)	Evaluate $\int \ell nx\,dx$	(iv)	Find $\int x\cos x\,dx$
(v)	Evaluate $\int e^x(\cos x + \sin x)\,dx$	(vi)	Evaluate $\int_{\pi/6}^{\pi/3} \cos t\,dt$
(vii)	Solve the differential equation $dy/dx = y/x^2$, $(y > 0)$		
(viii)	Find the midpoint of the line segment joining the two points $A(-8, 3)$ and $B(2, -1)$		
(ix)	Find an equation of the line through $A(-6, 5)$ having slope 7.		
(x)	Find the point of intersection of the lines. $x - 2y + 1 = 0$ and $2x - y + 2 = 0$		
(xi)	Transform the equation $5x - 12y + 39 = 0$ into two-intercept form.		
(xii)	Check whether the point $(-7, 6)$ lies above or below the line $4x + 3y - 9 = 0$		

4. Attempt any nine parts. $9 \times 2 = 18$

(i)	Graph the solution set of inequality $5x - 4y \le 20$	(ii)	Write the procedure for determining optimal solution.
(iii)	Find the centre and radius of the circle $5x^2 + 5y^2 + 24x + 36y + 10 = 0$		
(iv)	Find equation of circle with centre at $(5, -2)$ and radius 4.		
(v)	Check the position of point $(5, 6)$ with respect to circle $2x^2 + 2y^2 + 12x - 8y + 1 = 0$		
(vi)	Find vertex and directrix of the parabola $y^2 = -8(x - 3)$	(vii)	Find the eccentricity of the hyperbola $\frac{y^2}{16} - \frac{x^2}{49} = 1$
(viii)	Find the vector from the point A to the origin where $\overrightarrow{AB} = 4\underline{i} - 2\underline{j}$ and B is the point $(-2, 5)$		
(ix)	Find a vector whose magnitude is 2 and is parallel to $-\underline{i} + \underline{j} + \underline{k}$		
(x)	Determine whether the triple $30^o, 45^o, 60^o$ can be direction angle of the vector.		
(xi)	Find the angle between the vectors $\underline{u} = 2\underline{i} - \underline{j} + \underline{k}$ and $\underline{v} = -\underline{i} + \underline{j}$		
(xii)	Find α so that vectors $\underline{u} = 2\alpha\underline{i} + \underline{j} - \underline{k}$ and $\underline{v} = \underline{i} + \alpha\underline{j} + 4\underline{k}$ are perpendicular.		
(xiii)	If $\underline{a} \times \underline{b} = 0$ and $\underline{a} \cdot \underline{b} = 0$, what conclusion can be drawn about $\underline{a}$ or $\underline{b}$?		

SECTION-II

NOTE: Attempt any three questions. $3 \times 10 = 30$

5.(a)	Evaluate the limit $\lim_{\theta \to 0} \frac{1 - \cos p\theta}{1 - \cos q\theta}$	(b)	Differentiate $\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}$ w.r.t. "x"
6.(a)	If $x = \sin\theta$ and $y = sim(m\theta)$, show that $(1 - x^2)y_2 - xy_1 + m^2 y = 0$	(b)	Evaluate $\int x^3 e^{5x} dx$
7.(a)	Solve the differential equation $2x^2 y\frac{dy}{dx} = x^2 - 1$	(b)	Maximize $f(x, y) = 2x + 5y$ subject to the constraints $2y - x \le 8,\ x - y \le 4,\ x \ge 0,\ y \ge 0$
8.(a)	Find equations of two parallel lines perpendicular to $2x - y + 3 = 0$ and the product of $X-$ and $Y-$ intercepts of each is 3.		
(b)	Write equations of the tangents to the circle $x^2 + y^2 - 4x + 6y + 9 = 0$ at the points on the circle whose ordinate is -2.		
9.(a)	Find the centre, foci eccentricity and vertices of the given hyperbola $4y^2 + 12y - x^2 + 4x + 1 = 0$		
(b)	Prove that by vector method $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$		

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PAPER CODE: 4191	2026 (1st-A) INTERMEDIATE PART-II (12th Class)	Roll No: _____
MATHEMATICS PAPER-II GROUP-I		
TIME ALLOWED: 30 Minutes	OBJECTIVE	MAXIMUM MARKS: 20

Q.No.1 You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

S.#	QUESTIONS	A	B	C	D
1	If $f(x)=\sqrt{x+5}$ then $f(-6)$ is:	-1	1	0	Not defined
2	$\sec h^2 x =$	$1+\tanh^2 x$	$1-\tanh^2 x$	$\tanh^2 x-1$	$\coth^2 x-1$
3	If $f(x)=(x^3+1)^6$ then $f'(-2)$ is:	1210104	-1210104	1012104	-1012104
4	$\frac{d}{dx}(\sec x)$ at $x=\pi/4$ is:	$-\sqrt{2}$	$\frac{1}{\sqrt{2}}$	$\sqrt{2}$	$\frac{-1}{\sqrt{2}}$
5	If $y=\sin x$ then:	$y''+y=0$	$y''=0$	$y''=1$	$y''-y=0$
6	The stationary values of $f(x)=x^2-x-2$ is:	$-1/2$	$1/2$	2	-2
7	$\frac{d}{dx}\left(\cos^{-1} x/a\right)=$	$\frac{a}{\sqrt{a^2-x^2}}$	$\frac{-1}{\sqrt{1-x^2}}$	$\frac{-a}{\sqrt{a^2-x^2}}$	$\frac{-1}{\sqrt{a^2-x^2}}$
8	$\int(\cot x)\,dx =$	$\ell n\lvert\sec x\rvert$	$\ell n\lvert\sin x\rvert$	$\ell n\lvert\operatorname{cosec} x\rvert$	$\ell n\lvert\cos x\rvert$
9	$\int(x^2+1)x\,dx =$	$\frac{(x^2+1)^2}{4}$	$\frac{(x^2+1)^2}{2}$	$(x^2+1)^2$	$\frac{x^2+1}{4}$
10	$\int \ell nx\,dx =$	$x\ell nx+x$	$\ell nx+x$	$x\ell nx-x$	$\ell nx-x$
11	$\int_0^{\pi/4}\cos x\,dx =$	$\frac{-1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$\frac{-1}{\sqrt{2}}+1$	$\frac{-1}{\sqrt{2}}-1$
12	The slope of the line perpendicular to $3x-4y-3=0$ is:	$3/4$	$-3/4$	$-4/3$	$4/3$
13	The point of intersection of the three medians of a triangle is:	Centroid	Circum centre	In-centre	Orthocenter
14	The point (5, 8) lies:	On the line $2x-3y+6=0$	Below the line $2x-3y+6=0$	Above the line $2x-3y+6=0$	Both B and C
15	$x^2=4ay$ is the equation of:	Parabola	Circle	Ellipse	Hyperbola
16	The equation of the tangent to the circle $x^2+y^2=25$ at $(5\cos\theta, 5\sin\theta)$ is:	$5x+5y=25$	$5x\sin\theta+5y\cos\theta=25$	$5x\cos\theta+5y\sin\theta=25$	$x+y=25$
17	The vertex of the parabola $y^2=-8(x-3)$ is:	(0, 3)	(3, 0)	(−3, 0)	(0, −3)
18	The covertices of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with $b>a$:	$(\pm a, 0)$	$(0, \pm b)$	(a, b)	$(-a, -b)$
19	The direction cosines of $\underline{v}=\underline{i}-3\underline{j}+4\underline{k}$ are:	$\frac{1}{\sqrt{26}}, \frac{3}{\sqrt{26}}, \frac{4}{\sqrt{26}}$	$\frac{-1}{\sqrt{26}}, \frac{-3}{\sqrt{26}}, \frac{4}{\sqrt{26}}$	$\frac{1}{\sqrt{26}}, \frac{-3}{\sqrt{26}}, \frac{4}{\sqrt{26}}$	$\frac{-1}{\sqrt{26}}, \frac{3}{\sqrt{26}}, \frac{4}{\sqrt{26}}$
20	The value of $3j \,.\, k \times i$ is:	3	-3	0	1

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INTERMEDIATE PART-II (12th Class)	2026 (1st-A)	Roll No:
MATHEMATICS PAPER-II GROUP-II		
TIME ALLOWED: 2.30 Hours	SUBJECTIVE	MAXIMUM MARKS: 80

NOTE: Write same question number and its parts number on answer book, as given in the question paper.

SECTION-I

2. Attempt any eight parts. **8 × 2 = 16**

(i)	Define even and odd function.	(ii)	Find domain and range of f, if $f(x)=\sqrt{x^2-9}$
(iii)	Prove that $\cos h^2 x - \sin h^2 x = 1$	(iv)	Evaluate $\lim\limits_{x\to 0}\dfrac{\sin x^o}{x}$
(v)	Express in term of e :- $\lim\limits_{x\to+\infty}\left(1-\dfrac{1}{n}\right)^n$	(vi)	Find y_2 if $y=\sin 3x$
(vii)	Find $\dfrac{dy}{dx}$, if $y=(\ell n\tan hx)$	(viii)	Find $f'(x)$ if $f(x)=\ell n(e^x+e^{-x})$
(ix)	Differentiate $y=a^x$, w.r.t $'x'$	(x)	Find $\dfrac{dy}{dx}$ if $x=y\sin y$
(xi)	Prove that $\dfrac{dy}{dx}=\dfrac{1}{1+x^2}$, if $y=\tan^{-1}x$	(xii)	Differentiate $x^4+2x^3+x^2$, w.r.t $'x'$

3. Attempt any eight parts. **8 × 2 = 16**

(i)	Using differentials, find $\dfrac{dy}{dx}$ if $xy+x=4$	(ii)	Evaluate $\int\dfrac{3x+2}{\sqrt{x}}dx$
(iii)	Evaluate $\int\tan^2 x\,dx$	(iv)	Evaluate $\int\dfrac{x+2}{\sqrt{x+3}}dx$
(v)	Evaluate the definite integral $\int_1^2(x^2+1)\,dx$	(vi)	Find the area bounded by the curve $y=x^3+3x^2$ and the $x-$axis from $x=-3$ to $x=0$
(vii)	Solve the differential equation $\dfrac{dy}{dx}=-y$	(viii)	Find the point three fifth of the way along the line segment from $A(-5,8)$ to $B(5,3)$
(ix)	The $xy-$coordinate axes are rotated about the origin through the indicated angle. The new axes are $0X$ and $0Y$. Find the $XY-$coordinates of the point P with the given $xy-$coordinates $P(5,3)$; $\theta=45^o$.		
(x)	Check whether the two lines $2x+y-3=0$ and $4x+2y+5=0$ are parallel or perpendicular.		
(xi)	Show that the lines $4x-3y-8=0$, $3x-4y-6=0$ and $x-y-2=0$ are concurrent.		
(xii)	Find the pair of straight lines represented by $2x^2+3xy-5y^2=0$		

4. Attempt any nine parts. **9 × 2 = 18**

(i)	Define the linear inequality.	(ii)	Graph the solution set of the following linear inequality in $xy-$plane $2x+1\geq 0$
(iii)	Find an equation of the circle with centre at $(5,-2)$ and radius 4.		
(iv)	Find the centre and radius of the circle $x^2+y^2+12x-10y=0$		
(v)	Determine whether the point $P(-5,6)$ lies outside, on or inside the circle $x^2+y^2+4x-6y-12=0$		
(vi)	Find the focus and vertex of the parabola $y^2=8x$		
(vii)	Find an equation of the ellipse with vertices $(0,\pm 5)$, eccentricity $\dfrac{3}{5}$		
(viii)	Find the sum of the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ given the four points $A(1,-1)$, $B(2,0)$, $C(-1,3)$ and $D(-2,2)$		
(ix)	Find a vector whose magnitude is 4 and is parallel to $2\underline{i}-3\underline{j}+6\underline{k}$		
(x)	Find the constant a so that the vectors $\underline{v}=\underline{i}-3\underline{j}+4\underline{k}$ and $\underline{w}=a\underline{i}+9\underline{j}-12\underline{k}$ are parallel.		
(xi)	Calculate the projection of $\underline{a}$ along $\underline{b}$ when $\underline{a}=\underline{i}-\underline{k}$, $\underline{b}=\underline{j}+\underline{k}$		
(xii)	If $\underline{u}=2\underline{i}-\underline{j}+\underline{k}$ and $\underline{v}=4\underline{i}+2\underline{j}-\underline{k}$ then find the value of $\underline{v}\times\underline{u}$	(xiii)	Find the value of $3\underline{j}\,.\,\underline{k}\times\underline{i}$

SECTION-II

NOTE: Attempt any three questions. **3 × 10 = 30**

5.(a)	Discuss the continuity of $f(x)$ at $x=-2$ if $f(x)=\begin{cases}3x & \text{if } x\leq -2\\ x^2-1 & \text{if } -2<x<2\\ 3 & \text{if } x\geq 2\end{cases}$	(b)	Differentiate $\sec^{-1}\left(\dfrac{x^2+1}{x^2-1}\right)$ w.r.t. $'x'$
6.(a)	Differentiate $(\ell nx)^x$ w.r.t. $'x'$	(b)	Evaluate $\int\tan^3 x\sec x\,dx$
7.(a)	Find the area between the $x-$axis and the curve $y=\sqrt{2ax-x^2}$, where $a>0$		
(b)	Minimize $z=3x+y$ subject to the constraints $3x+5y\geq 15$; $x+3y\geq 9$, $x\geq 0$; $y\geq 0$		
8.(a)	Find the centre and radius of the circle with the equation $4x^2+4y^2-8x+12y-25=0$		
(b)	By means of slopes, show that the points $A(4,-5)$, $B(7,5)$ and $C(10,15)$ lie on the same line.		
9.(a)	Find the centre, foci, eccentricity and vertices of the ellipse. $x^2+16x+4y^2-16y+76=0$		
(b)	Prove that by vector method $\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$		

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MATHEMATICS PAPER-II GROUP-II		
TIME ALLOWED: 30 Minutes	OBJECTIVE	MAXIMUM MARKS: 20

Q.No.1 You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

S.#	QUESTIONS	A	B	C	D
1	If $f(x)=\sin x+\cos x$ then $f(\pi/4)=$	$\sqrt{2}$	$\frac{1}{\sqrt{2}}$	$-\sqrt{2}$	$\frac{-1}{\sqrt{2}}$
2	If $f(x)=\frac{1}{x+3}$, $x\neq -3$ then range of $f^{-1}(x)$ is:	R	$R-\{3\}$	$R-\{-3\}$	$R-\{1\}$
3	$\frac{d}{dx}(\sqrt{x})=$	$\frac{1}{\sqrt{x}}$	$\frac{1}{2\sqrt{x}}$	$\frac{-1}{2\sqrt{x}}$	$\frac{-1}{\sqrt{x}}$
4	If $y=e^{-ax}$ then $\frac{d^3y}{dx^3}$ is:	$-a^3y$	a^3y	a^2y	$-a^2y$
5	$\frac{d}{dx}(\cot h^{-1}x)$ is:	$\frac{1}{1+x^2}$	$\frac{-1}{1+x^2}$	$\frac{1}{1-x^2}$	$\frac{-1}{1-x^2}$
6	If $y=e^{3x}$ then y_3 is:	$27e^{3x}$	$9e^{3x}$	$3e^{3x}$	e^{3x}
7	$\int \sin 5x\,dx=$	$-\cos 5x$	$\cos 5x$	$\frac{-1}{5}\cos 5x$	$\frac{1}{5}\cos 5x$
8	$\int \frac{1}{x+2}\,dx=$	ℓnx	$\ell n(x+2)$	$\frac{1}{(x+2)^2}$	$\frac{-1}{(x+2)^2}$
9	$\int (x^2-1)^{1/2}x\,dx=$	$\frac{1}{3}(x^2+1)^{3/2}$	$\frac{1}{2}(x^2+1)^{3/2}$	$\frac{-1}{2}(x^2+1)^{3/2}$	$\frac{-1}{3}(x^2+1)^{3/2}$
10	$\int \frac{1}{2x-1}\,dx=$	$\ell n(2x-1)$	$\frac{1}{2}\ell n(2x-1)$	$\frac{-1}{2}\ell n(2x-1)$	$-\ell n(2x-1)$
11	The slop of the line parallel to $3x+4y+7=0$ is:	$3/4$	$-3/4$	$4/3$	$-4/3$
12	The vertical line through $(-3, 6)$ is:	$x=6$	$x=-3$	$x=-6$	$x=3$
13	The distance of the point $(3, -7)$ from $x-axis$ is:	3	-3	-7	7
14	The solution of $x+2y<6$ is:	(1, 1)	(1, 3)	(1, 4)	(1, 5)
15	$x^2=4ay$ is the equation of:	Ellipse	Circle	Parabola	Hyperbola
16	In hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$	$b^2=a^2+c^2$	$b^2=c^2-a^2$	$b^2=a^2-c^2$	$b^2=a^2$
17	An parabola $y^2=4ax$	$e=1$	$e<1$	$e>1$	$e=0$
18	If $\underline{a}$ & $\underline{b}$ have the opposite direction then $\underline{a}\cdot\underline{b}=$	ab	$-ab$	$ab\sin\theta$	$ab\tan\theta$
19	If $\frac{\underline{a}\cdot\underline{b}}{ab}=-1/2$ then θ is:	$\frac{5\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$	$\frac{2\pi}{3}$
20	For three coplaner vectors $\underline{a}$, $\underline{b}$ & $\underline{c}$ $\underline{a}\cdot\underline{b}\times c=$	2	1	-1	0

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